

SEMESTER-III

UMT306: SIGNAL CONDITIONING AND DATA ACQUISITION				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To understand concepts of signal conditioning and acquiring the data from transducers/input devices and their interfacing to build a complete mechatronics system.				
Syllabus Signal conditioning circuits: Introduction to Operational amplifiers, input-output characteristics, frequency response, Slew rate. Op-Amp based signal conditioning circuits, Integrator, differentiator, log amplifier, Difference amplifier, Instrumentation amplifier, AD620. Basic Bridge amplifier and its use with strain gauge and temperature sensors. DC Bridges, use of Wheatstone bridge for resistive transducers, voltage and current sensitivity of dc bridge. Active Filters in instrumentation circuits: High pass low pass, Band pass, Design of Butterworth and Chebyshev filters. Data Acquisition Techniques: Analog and digital data acquisition, Sensor/Transducer interfacing, unipolar and bipolar transducers, Sample and hold circuits, Sampling and quantization, Interference, Grounding and Shielding. Analog to Digital convertors (ADC): ramp type and SAR, and Digital to Analog Convertors (DAC) R-2R based DAC, Multiplexer and de-multiplexer. Data Transfer Techniques: Serial data transmission methods and standards RS 232-C: specifications, connection and timing diagram, MAX232, RS422 and RS485, 4-20 mA current loop, HART protocol, Specification, connections and techniques of GPIB/IEEE-488, Universal Serial Bus (USB), Bluetooth protocols, IR receiver and transmitters for remote control applications. Data Acquisition System (DAS): Basic building blocks of DAS, types and configurations. Single channel and multichannel, Design of data acquisition system, Data acquisition cards and Remote Terminal Units, PC-Based data acquisition system: LabVIEW based data acquisition. Supervisory Control and data acquisition system (SCADA).				
Laboratory Work Introduction to PSPICE/OrCAD/Multisim for circuit simulation, Op-amp as a comparator and its application, Integrator and differentiator, Log antilog amplifier, Simulation of the above applications in OrCAD or MultiSim. Experiments related to Instrumentation Amplifier, High pass, low pass and Band pass filters. PC based Data acquisition using LabVIEW/DaisyLab/Matlab, MyDAQ based experiment				
Course Learning Outcomes (CLOs) The students will be able to: 1. elucidate signal conditioning circuits 2. design basic signal conditioning circuits 3. explain various data transfer techniques 4. explain the elements of data acquisition techniques. 5. elucidate the components of data acquisition system				
Text Books 1. Operational Amplifiers and Linear Integrated Circuits, Coughlin, R.F., Pearson				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

- Education, 2006.
- Electronic Instrumentation, Kalsi, H.S., Tata McGraw Hill, 2002.
 - Op-Amp and Linear Integrated Circuits, Gayakwad, R.A., Pearson Education, 2002.
 - Data Acquisition Using LabVIEW, Behzad Ehsani, Packt, 2016
 - Microprocessor PC Hardware and Interfacing, Mathivanan, N., Prentice Hall of India Private Limited, 2007.

Reference Books

- Electronic Instruments and Instrumentation Technology, Ananad, M.M.S., Prentice Hall of India Private Limited, 2004.
- Transducers and Instrumentation, Murthy, D.V.S., Prentice Hall of India Private Limited, 2006.

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	35

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